

Energy  
EfficiencyThroughput  
per Watt

Low Latency

Scalability

Sustainability

**System Software / Orchestration**

Architectural-Aware Scheduling

Dynamic Workload Mapping

Latency / QoS Allocation

Control Decisions

Telemetry Feedback

**Runtime Power and Adaptation Controls**

DVFS

Power Capping

Fine-Grained Power  
Capping

Runtime Telemetry

Runtime Control Commands

Energy / Performance Feedback

**Accelerator Architecture**

GPUs

Processing-in-  
Memory (PIM)Coarse-Grained  
Linear Arrays  
(CGLA)Wafer-Scale  
AcceleratorsReconfigurable  
GPU Arrays

Placement And Execution Demands

Utilization Feedback

**Memory and Communication Substrate**HBM / On-Chip  
CachesCompute-Memory  
Co-LocationMinimized Off-Chip  
CommunicationMesh- and  
Interconnect- Aware  
Placement

Cross - Layer Coordination